



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.



Session 2026-2027

Vision: Dream of where you want.	Mission: Means to achieve Vision
---	---

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

(Signature and Date in Handwritten)



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session	2026-27	Course Name	Web Technology Lab
Semester	3	Course Code	23CT1301
Roll No	13	Name of Student	Juhi Parwani

Practical Number	1(A)
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	To develop a simple calculator using HTML and JavaScript and perform basic arithmetic operations such as addition, subtraction, multiplication, and division using DOM manipulation and event handling.
Problem Definition	Develop a calculator application that accepts numbers from the user and performs basic arithmetic operations. The calculator should provide buttons for numbers, arithmetic operators, equals, and clear operations. JavaScript event listeners are used to handle button clicks and keyboard inputs.
Theory (100 words)	JavaScript is a scripting language used to make web pages interactive and dynamic. DOM manipulation allows JavaScript to access and modify HTML elements. Event handling is used to perform actions when the user clicks a button or presses a keyboard key. Functions are used to organize the calculator operations such as adding numbers, selecting an operator, calculating the result, and clearing the display



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Procedure and Execution

(100 Words)

Step for Implementation:

1. Create a project folder in VS Code.
2. Create `index.html` and `script.js`.
3. Design the calculator buttons and display using HTML.
4. Connect the JavaScript file with the HTML file.
5. Use `getElementById()` to access calculator elements.
6. Create functions for entering numbers and selecting operators.
7. Perform addition, subtraction, multiplication, and division.
8. Add event listeners to calculator buttons.
9. Add keyboard event handling.
10. Run the application using a web browser.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Calculator</title>
</head>
<body>

  <h2>Simple Calculator</h2>

  <input type="text" id="display" readonly>

  <br><br>

  <button id="seven">7</button>
  <button id="eight">8</button>
  <button id="nine">9</button>
  <button id="divide">/</button>

  <br><br>

  <button id="four">4</button>
  <button id="five">5</button>
  <button id="six">6</button>
```



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.



```
<button id="multiply">*</button>
```

```
<br><br>
```

```
<button id="one">1</button>
```

```
<button id="two">2</button>
```

```
<button id="three">3</button>
```

```
<button id="subtract">-</button>
```

```
<br><br>
```

```
<button id="zero">0</button>
```

```
<button id="clear">C</button>
```

```
<button id="equals">=</button>
```

```
<button id="add">+</button>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

```
let display = document.getElementById("display");
```

```
let firstNumber = "";
```

```
let operator = "";
```

```
function addNumber(number) {  
    display.value += number;  
}
```

```
function setOperator(op) {  
    firstNumber = display.value;  
    operator = op;  
    display.value = "";  
}
```



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

```
function calculate() {
    let secondNumber = display.value;
    let result;

    if (operator == "+") {
        result = Number(firstNumber) +
Number(secondNumber);
    }
    else if (operator == "-") {
        result = Number(firstNumber) -
Number(secondNumber);
    }
    else if (operator == "*") {
        result = Number(firstNumber) *
Number(secondNumber);
    }
    else if (operator == "/") {
        result = Number(firstNumber) /
Number(secondNumber);
    }

    display.value = result;
}
function clearDisplay() {
    display.value = "";
    firstNumber = "";
    operator = "";
}

document.getElementById("seven").addEventListener("click"
, function() {
    addNumber(7);
});
```



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

```
document.getElementById("eight").addEventListener("click",
, function() {
    addNumber(8);
});

document.getElementById("nine").addEventListener("click",
function() {
    addNumber(9);
});

document.getElementById("four").addEventListener("click",
function() {
    addNumber(4);
});

document.getElementById("five").addEventListener("click",
function() {
    addNumber(5);
});

document.getElementById("six").addEventListener("click",
function() {
    addNumber(6);
});

document.getElementById("one").addEventListener("click",
function() {
    addNumber(1);
});

document.getElementById("two").addEventListener("click",
function() {
    addNumber(2);
});

document.getElementById("three").addEventListener("click",
, function() {
```



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

```
        addNumber(3);
    });

    document.getElementById("zero").addEventListener("click",
function() {
    addNumber(0);
});

    document.getElementById("add").addEventListener("click",
function() {
    setOperator("+");
});

    document.getElementById("subtract").addEventListener("cli
ck", function() {
    setOperator("-");
});

    document.getElementById("multiply").addEventListener("cli
ck", function() {
    setOperator("*");
});

    document.getElementById("divide").addEventListener("click
", function() {
    setOperator("/");
});

    document.getElementById("equals").addEventListener("click
", function() {
    calculate();
});

    document.getElementById("clear").addEventListener("click"
, function() {
    clearDisplay();
});
```



Department of Computer Technology

Vision Statement of the Department.

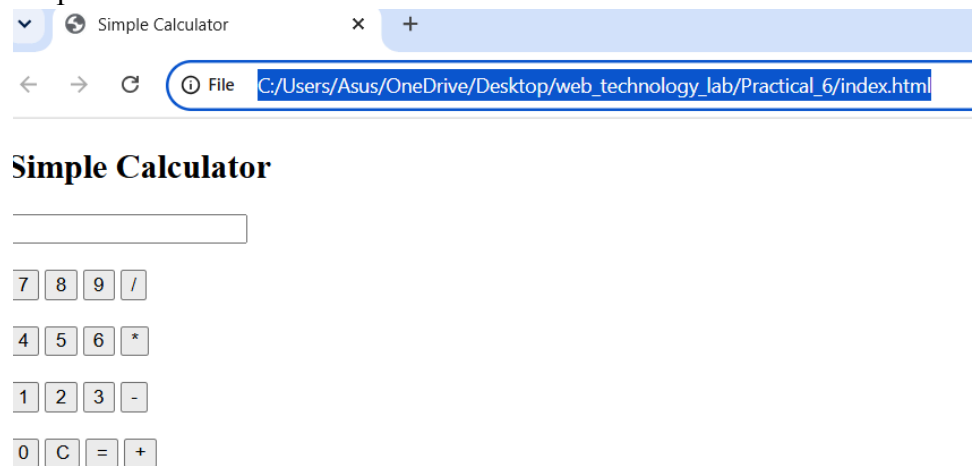
To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

```
document.addEventListener("keydown", function(event) {  
  
    if (event.key >= "0" && event.key <= "9") {  
        addNumber(event.key);  
    }  
  
    if (event.key == "Enter") {  
        calculate();  
    }  
  
    if (event.key == "Escape") {  
        clearDisplay();  
    }  
  
});
```

Output:



Output Analysis

The calculator successfully accepts numbers from the user and performs basic arithmetic operations. Button click events are handled using JavaScript event listeners. The result is displayed dynamically in the calculator display. Keyboard events are also handled for number input, Enter, and Escape keys.



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Link of student Github profile where lab assignment has been uploaded	https://github.com/juhiparwa-1507/Web-Technology-Lab.git
Conclusion	The calculator application was successfully developed using HTML and JavaScript. The practical demonstrates DOM manipulation, functions, conditional statements, event handling, and keyboard events to perform basic arithmetic operations.
Plag Report (Similarity index < 12%)	Plagiarism Report: To be attached.
Date	28-09-26